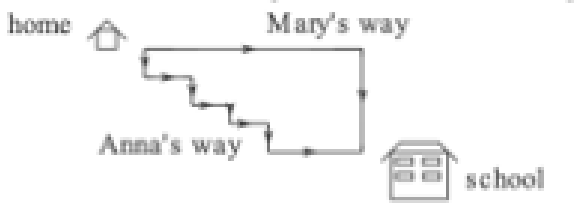
3 Points Each

1999

1. 1999 – 999 – 99 – 9 =

(A) 1900
(B) 1090
(C) 1000
(D) 1990
(E) 1009

2. Anna and her sister Mary take two different ways to school. Whose way is longer?

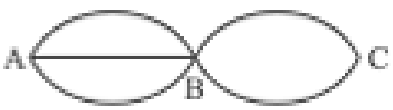


(A) Anna's way
(B) Mary's way
(C) It depends on the distance to the school.
(D) Both ways have the same length.
(E) It is impossible to determine.

3. One quarter of one half of double 32 is equal to

(A) 4
(B) 8
(C) 16
(D) 32
(E) 64

4. How many different ways connect city A to city C?

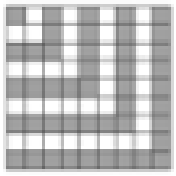


(A) 2
(B) 3
(C) 5
(D) 6
(E) 9

5. How many times faster does the minute hand of any clock move than the hour hand?

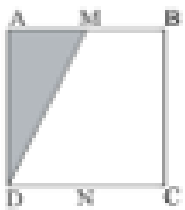
(A) 6
(B) 12
(C) 9
(D) 10
(E) 15

6. In the picture to the right some small squares are shaded. The big square has dimensions of 9 x 9 and is made up of both shaded and white small squares. Find the difference between the number of shaded squares and white squares.



(A) 0
(B) 1
(C) 5
(D) 9
(E) 10

7. ABCD is a square and point M is the midpoint of side AB. The shaded part has an area equal to 9 cm² (see the picture). What is the area of square ABCD?



(A) 18 cm²
(B) 27 cm²
(C) 32 cm²
(D) 36 cm²
(E) 45 cm²

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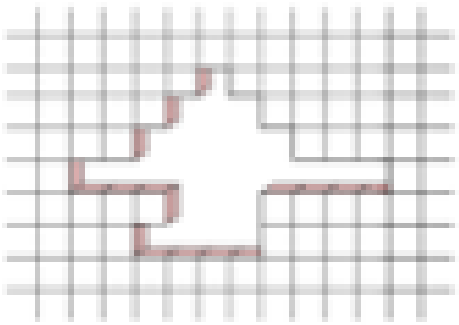
8. A movie started at 1:47 p.m. and finished at 4:18 p.m. How long was the movie?

(A) 185 min
(B) 151 min
(C) 91 min
(D) 149 min
(E) 209 min

9. At a birthday party, one person found out that no two of the people present were born in the same month. What is the greatest possible number of people at that party?

(A) 11
(B) 12
(C) 13
(D) 24
(E) 44

10. How many bricks were taken out from



(A) 26
(B) 32
(C) 36
(D) 40
(E) 42

4 Points Each

11. John opened a book and found that the sum of the page number on the left and the page number on the right is equal to 25. What is the product of the two page numbers?

(A) 169
(B) 144
(C) 150
(D) 156
(E) 1998

12. A certain kangaroo's jump is 5 m long. How many jumps does it need to make to cover a distance of 5000m + 5000 dm + 5000 cm + 5000 mm?

(A) 1000
(B) 1100
(C) 1110
(D) 1111
(E) 5555

13. A stick that in reality measures 1 m is 2 cm long in a certain picture, and in that same picture the height of a fence is 4.5 cm. What is the actual height of the fence in cm?

(A) 450
(B) 225
(C) 45
(D) 22.5
(E) 4.5

14. Find the sum of the missing digits.

(A) 6
(B) 8
(C) 10
(D) 12
(E) 14

$$\begin{array}{r} 6A3 \\ \times \quad 5 \\ \hline 346B \end{array}$$

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15. Which of the figures below will form a rectangle when combined with the figure to the right?



(A) 
(B) 
(C) 
(D) 
(E) 

16. The dog is 9 times as heavy as the cat, the mouse is 20 times lighter than the cat, and the turnip is 6 times as heavy as the mouse. How many times is the dog as heavy as the turnip?

(A) 30
(B) 27
(C) 1080
(D) 15
(E) The dog is lighter than the turnip.

17. A kangaroo wants to make a rectangular quilt 1.5 m long and 1 m wide using square scraps which measure 10 cm x 10 cm. At every point where four squares meet she wants to place a fancy button. How many buttons will she need?

(A) 150
(B) 104
(C) 126
(D) 140
(E) 135

18. One bowl contained 26 liters of water and another bowl contained 7 liters of water. The same amount of water was added to each bowl, and now the second bowl contains 3 times less water than the first bowl. How many liters of water were added to each bowl?

(A) 2.5
(B) 5
(C) 7.5
(D) 10
(E) 15

19. In a magic square, the sum of the numbers in the cells in each vertical column, in each horizontal row, and along each diagonal is the same. The square shown to the right is a magic square. Two of the numbers have been taken out, and three were covered with the letters A, B, and C. Find the sum A+B+ C.

16	3	A
C	10	
B		4

(A) 30
(B) 41
(C) 14
(D) 25
(E) It is impossible to determine.

20. John and Adam are making a square using small squares with the same dimensions. Adam places one red square down, and John adds 8 green squares around it to make a second square. Then Adam places 16 yellow squares around this square to make a third square, and so forth. How many squares does Adam have to place to make the fifth square?

(A) 32
(B) 64
(C) 81
(D) 121
(E) 125

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5 Points Each

21. Ela came to Anna's birthday party 5 minutes earlier than Stan but 3 minutes later than Iwona. Iwona left first. She left 2 minutes earlier than Stan and 5 minutes earlier than Ela. How many minutes longer was Ela at the party than Stan?

(A) 2
(B) 4
(C) 6
(D) 8
(E) Stan stayed longer than Ela.

22. Six hundred and six people can eat six hundred and six hotdogs, six hundred of which are with mustard and six are without mustard. How many hotdogs without mustard will you need for six hundred and six thousand six hundred and six people?

(A) 606
(B) 1000
(C) 6006
(D) 606,606
(E) 600,600

23. Four squirrels ate 1999 nuts altogether, and each one ate at least 100 nuts. The first squirrel ate more nuts than any other squirrel. The second and third squirrels together ate 1265 nuts. How many nuts did the first squirrel eat?

(A) 598
(B) 271
(C) 629
(D) 634
(E) other answer

24. When it is raining, the cat stays in the room or in the basement. When the cat stays in the room, the mouse is in the foyer and the cheese is in the refrigerator. When the cheese is on the table and the cat stays in the basement, the mouse is in the room. Right now, it is raining and the cheese is on the table. So, for sure:

(A) The cat is in the room.
(B) The cat is in the room and the mouse is in the foyer.
(C) The mouse is in the foyer.
(D) The cat is in the basement and the mouse is in the room.
(E) This situation is impossible.

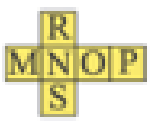
25. What is the maximum number of acute angles that can be made by six rays coming out from the same point?

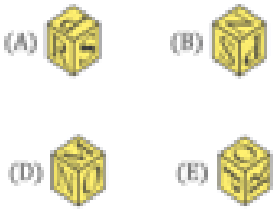
(A) 6
(B) 8
(C) 9
(D) 12
(E) 15

26. The "yield" of 36 equals 18, the "yield" of 325 equals 30, the "yield" of 45 equals 20, and the "yield" of 30 equals 0. What is the "yield" of 531?

(A) 10
(B) 15
(C) 16
(D) 21
(E) 22

27. The diagram to the right corresponds to only one of the cubes below. Which one?





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28. In each one of the 5 cups shown below you can find coffee, cocoa, or tea. There is twice as much coffee as cocoa. None of the beverages was poured into three cups. In which cup- is the cocoa?



29. How many squares are there in the figure to the right?



(A) 46
(B) 47
(C) 45
(D) 33
(E) 37

30. An electronic clock shows hours, minutes, and seconds. Right now it shows 19:58:47. As you can see, all the digits are different. After how many seconds will you see all different digits again?

(A) 40
(B) 73
(C) 156
(D) 157
(E) 898

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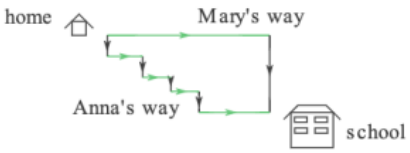
Answers key:

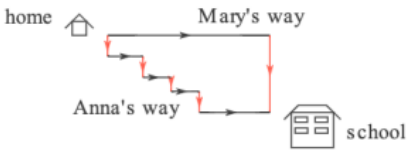
1. B
2. D
3. B
4. D
5. B
6. D
7. D
8. B
9. B
10. A
11. D
12. D
13. B
14. E
15. C
16. A
17. C
18. A
19. D
20. A
21. D
22. C
23. D
24. D
25. E
26. B
27. E
28. B
29. B
30. E

for purchaser's use only

1. (B) 1090
Do the operations left to right.
$1999 - 999 + 99 - 9 =$
$= 1000 + 99 - 9 =$
$= 1099 - 9 =$
$= 1090$

2. (D) Both ways have the same length.





Ann's horizontal segments (red) match Mary's horizontal segment (green).

Ann's vertical segments (red) match Mary's vertical segment (green).

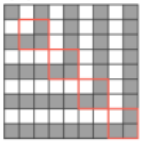
Therefore, both ways have the same length.

3. (B) 8
$\frac{1}{4} \times \left[\frac{1}{2} \times (2 \times 32)\right] = \frac{1}{4} \times 32 = 8$

4. (D) 6
There are 3 ways from A to B (the left upper arc, the left straight segment, and the left lower arc) and 2 ways from B to C (the right upper arc and the right lower arc), so $3 \times 2 = 6$ different ways that connect city A to city C (any left part can be connected with any right part).

5. (B) 12
The minute hand makes one full revolution during 1 hour and the hour hand makes one full revolution during 12 hours, so the minute hand moves 12 times faster than the hour hand.

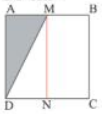
6. (D) 9



To the left of each red frame the shaded unit squares match the white unit squares. The same is true for the unit squares above each red frame, so without the upper left unit square (shaded) and without unit squares inside red frames the shaded unit squares match the white unit squares. In each red frame there are 2 more shaded squares than white squares, so in the 4 red frames there are $4 \times 2 = 8$ more shaded squares than white squares. Add the upper left shaded unit square to see that $4 \times 2 + 1 = 8 + 1 = 9$ is the difference between the number of shaded squares and white squares.

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7. (D) 36 cm²



MN is perpendicular to AB and DC and M is the midpoint of AB, so the shaded area is $\frac{1}{4}$ of the square ABCD. The shaded part has an area equal to 9 cm², so the area of the square ABCD is 4 × 9 cm² = 36 cm².

8. (B) 151 min

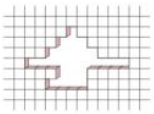
The time from 1:47 p.m. to 4:18 p.m. is 13 min + 2 hours + 18 min = 2 h 31 min, which is 2 × 60 min + 31 min or 151 min, so the movie was 151 min long.

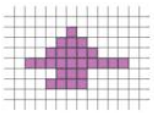
9. (B) 12

There are 12 different months, so the greatest possible number of people at the party is 12, just one birthday each month.

10. (A) 26

The empty spaces are filled with same size pink bricks. Count them row by row to see that there are 1 + 3 + 5 + 10 + 3 + 4 = 26 pink bricks, so 26 bricks were taken from the wall.





11. (D) 156

The numbers on the two pages of the book are consecutive natural numbers. Since their sum is 25, the numbers are 12 and 13. Their product is 12 × 13 = 156.

12. (D) 1111

1 m = 10 dm = 100 cm = 1000 mm

5000 m + 5000 dm + 5000 cm + 5000 mm = 5000 m + 500 m + 50 m + 5 m = 5555 m. The kangaroo's jump is 5 m long and 5555 ÷ 5 = 1111, so the kangaroo needs 1111 jumps to cover the distance of 5000 m + 5000 dm + 5000 cm + 5000 mm.

13. (B) 225

1 m in reality is represented by 2 cm in the picture.

4.5 cm = 2.25 × 2 cm, so the actual height of the fence is 2.25 × 1 m = 2.25 m = 225 cm.

14. (E) 14

$$\begin{array}{r} 6A3 \\ \times\quad 5 \\ \hline 346B \end{array}$$

The given product ends with 5 since 3 × 5 = 15 ends with 5, so the product is 3465.

The number 5 is one of the factors, so the other factor is 3465 ÷ 5 = 693.

The missing digits are 9 and 5 and their sum is 9 + 5 = 14.

15. (C)



As you can see in the picture below, only figure C will form a rectangle when combined with the figure on the right.



(A)  (B)  (C)  (D)  (E) 

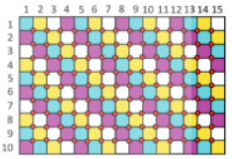
16. (A) 30

The dog's weight is 9 times the cat's weight, which is 20 times the mouse's weight, so the dog's weight is 180 times the mouse's weight since 9 × 20 = 180. 180 = 30 × 6 and the turnip's weight is 6 times the mouse's weight, so the dog is 30 times heavier than the turnip.

17. (C) 126

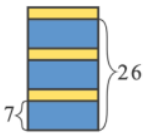
1.5 m = 150 cm = 15 × 10 cm and 1 m = 100 cm = 10 × 10 cm, so the quilt will have 15 columns and 10 rows of square scraps.

The numbers of rows and columns where the squares meet (the seams) will be one less than the numbers of columns and rows of squares. Thus, there will be 14 columns of buttons with 9 buttons in each column, so the kangaroo will need 9 × 14 = 126 buttons.



18. (A) 2.5

Each blue strip represents 7 liters of water, and each yellow strip represents the amount of water added to the bowl with 7 liters of water. It is the same as the amount of water added to the bowl with 26 liters of water. The bowl with 7 liters of water and the additional amount of water is shown 3 times and it is the same as 26 liters of water with the additional amount of water. Thus, 3 blue strips and 2 yellow strips represent 26 liters of water. Hence, 1 yellow strip represents (26 – 3 × 7) ÷ 2 = 5 ÷ 2 = 2.5 liters of water. Therefore, 2.5 liters of water was added to each bowl.



19. (D) 25

16	3	A
C	10	
B		4

The magic square shown to the left has only one solution. The sum of numbers along one diagonal is 16 + 10 + 4 = 30. This is the same sum as in each of the rows and each of the columns.

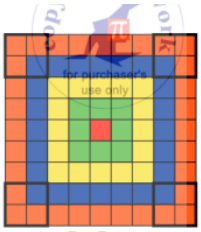
The number covered with A is 30 – (16 + 3) = 30 – 19 = 11. In the first column, the sum is 16 + B + C = 30, so B + C = 30 – 16 = 14. We don't need to solve further to find that A + B + C = 11 + 14 = 25.

16	3	11
5	10	15
9	17	4

If we do want to solve the whole square, the last entry in the 2nd column is 30 – (3 + 10) = 17. After that the middle entry in the 3rd column is computed as 30 – (11 + 4) = 15. Then, the first entries of the second and third row are computed as 30 – (10 + 15) = 5 and 30 – (17 + 4) = 9.

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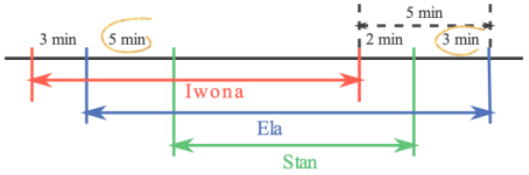
20. (A) 32



The number of squares is: 8 green, 16 yellow, 24 blue, and 32 orange. Thus, Adam needs to add 32 squares (the orange ones) to the fourth square to make the fifth one. Each time, the number of squares of the new color increases by 8 compared to the previous color. To see it, look at the four 2×2 corner squares with black perimeters. Without the unit squares inside them, the orange squares match the blue squares. In each of the four 2×2 squares there are 3 orange and 1 blue square, so there are $4 \times (3 - 1) = 8$ more orange squares than the blue ones.

Another solution: The fifth square is a 9×9 square and the fourth square is a 7×7 square, so the number of new unit squares (in orange) is $9^2 - 7^2 = 32$.

21. (D) 8



Ela arrived at the birthday party 5 minutes before Stan. Iwona left the party 2 minutes before Stan and 5 minutes before Ela, so Ela left the party 3 minutes after Stan. Hence, Ela was at the party $5 + 3 = 8$ minutes longer than Stan.

22. (C) 6006

For each group of 606 people we need 6 hotdogs without mustard.
$606{,}606 = 606{,}000 + 606 = 1000 \times 606 + 606 = 1001 \times 606$, so for 606,606 people we will need $1001 \times 6 = 6006$ hotdogs without mustard.

23. (D) 634

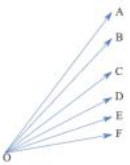
Either the second or the third squirrel ate at least 633 nuts. Otherwise, each of them would eat 632 nuts or less, so together they would eat not more than $632 + 632 = 1264$ nuts, which is less than the 1265 nuts actually eaten by the two squirrels. The first squirrel ate more nuts than any other squirrel, so the first squirrel ate at least 634 nuts. Together the first and the fourth squirrel ate $1999 - 1265 = 734$ nuts. The fourth squirrel alone ate 100 or more nuts, so the first squirrel ate not more than $734 - 100 = 634$ nuts but we already know that the first squirrel ate at least 634 nuts. Therefore, the first squirrel ate exactly 634 nuts. Consequently, the fourth squirrel ate exactly $734 - 634 = 100$ nuts. Of the other two squirrels, one ate exactly 633 nuts and another of them ate exactly 632 nuts.

24. (D) The cat is in the basement and the mouse is in the room.

Right now, the cheese is on the table, so the cat is not in the room because the cat in the room means that the cheese is in the refrigerator.

Thus, the cat is in the basement and the cheese is on the table. It means that the mouse is in the room. Therefore, the cat is in the basement and the mouse is in the room. This is the only option that is true. Each of the other options is false.

25. (E) 15



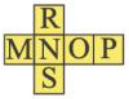
$\angle AOF$ and all other angles are acute angles (less than 90°), so we can pick any two rays to have an acute angle. Here is the list of all te angles.
$\angle AOB$, $\angle AOC$, $\angle AOD$, $\angle AOE$, $\angle AOF$, $\angle BOC$, $\angle BOD$, $\angle BOE$, $\angle BOF$, $\angle COD$, $\angle COE$, $\angle COF$, $\angle DOE$, $\angle DOF$, and $\angle EOF$. There are 15 such angles.
For any collection of 6 items, 15 is the number of all pairs that can be formed by using only these 6 items. If we call the item A, B, C, D, E, and F, then the possible pairs are: AB, AC, AD, AE, AF, BC, BD, BE, BF, CD, CE, CF, DE, DF, and EF.

26. (B) 15

It seems that the "yield" for any natural number written in the decimal system is the product of its digits, so yield $(30) = 3 \times 0 = 0$, yield $(36) = 3 \times 6 = 18$, yield $(45) = 4 \times 5 = 20$, and yield $(325) = 3 \times 2 \times 5 = 30$. If this is the case, then yield $(531) = 5 \times 3 \times 1 = 15$.

27. (E) 

When folding the net into a cube, keep the face with N as the front face, the face with M as the left face, the face with R as the top face, the face with S as the bottom face, the face with O as the right face, and the face with P as the back face. The faces with R and S, the faces with N and P, and the faces with M and O are pairs of opposite faces which eliminates



cubes (A) 

, (B) 

, and (D) 

as cubes of the net.

In (C) 

the positions of the letters M and N with respect to the common edge are incorrect. Two parts (legs) of M and two parts (legs) of N are perpendicular to the common edge of the two faces but in the net these parts are parallel to the common edge. This eliminates cube (C) as a cube of the net. (E) is the only option left.

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28. (B) 

Each cup contains only one beverage, so there is exactly one beverage that is only in 1 cup and each of the other two beverages is in 2 cups (1 + 2 + 2 = 5). The problem asks which cup the cocoa is in, implying that the cocoa is the beverage that is in one cup. If this is the case, the tea and coffee are each in two cups.

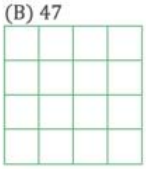
The one cup of cocoa must match half of the coffee in two other cups. From 5 given cups we can select two cups in 10 ways. Add the numbers of the two selected cups and divide the sum by 2. If the result matches a number of another cup, then the cup matched contains cocoa.

(950 + 750) ÷ 2 = 850; (950 + 550) ÷ 2 = 750; (950 + 475) ÷ 2 = 712.5;
(950 + 325) ÷ 2 = 637.5; (750 + 550) ÷ 2 = 650; (750 + 475) ÷ 2 = 612.5;
(750 + 325) ÷ 2 = 537.5; (550 + 475) ÷ 2 = 512.5; (550 +325) ÷ 2 = 437.5;
and (475 + 325) ÷ 2 = 400.

Only selecting the 950 g and 550 g cups for coffee produces a match. The cocoa is in the 750 g cup, the coffee is in the 950 g and 550 g cups, and tea is in the 475 g and 325 g cups.

The amount of coffee is 1500 g, which is twice as much as the amount of cocoa.

29. (B) 47



In the figure to the left there are squares of sizes 1 × 1, 2 × 2, 3 × 3, and 4 × 4.

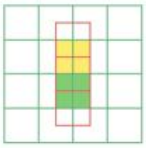
To get a square of the size 1 × 1 you need a pair of vertical lines 1 unit apart (there are 4 such pairs) and a pair of horizontal lines 1 unit apart (there are 4 such pairs). Thus, there are $4^2 = 16$ squares of the size 1 × 1.

To get a square of the size 2 × 2 you need a pair of vertical lines 2 units apart (there are 3 such pairs) and a pair of horizontal lines 2 units apart (there are 3 such pairs). Thus, there are $3^2 = 9$ squares of the size 2 × 2.

To get a square of the size 3 × 3 you need a pair of vertical lines 3 units apart (there are 2 such pairs) and a pair of horizontal lines 3 units apart (there are 2 such pairs). Thus, there are $2^2 = 4$ squares of the size 3 × 3.

To get a square of the size 4 × 4 you need a pair of vertical lines 4 units apart (there is 1 such a pair) and a pair of horizontal lines 4 units apart (there is 1 such a pair). Thus, there is $1^2 = 1$ square of the size 4 × 4.

Therefore, there are $1^2 + 2^2 + 3^2 + 4^2 = 30$ squares in the 4 × 4 square above.



3 squares (with red perimeters) are added to the above figure. In addition to these 3 squares, there are 2 more 1 × 1 new squares (shown in colors). There are also 2 × 6 = 12 very small new squares. Hence, 3 + 2 + 12 = 17 new squares have been added to the 30 squares. Altogether, there are 30 + 17 = 47 squares.

30. (E) 898

Right now the electronic clock shows 19:58:47. In one second it will show 19:58:48, which contains two 8s, and one second later it will show 19:58:49, which contains two 9s. Afterwards the digit 5 will be repeated multiple times and then 9 will be repeated for one minute. Once we reach 8 PM, 20:00:00, we will see 20:13:45 as the first electronic display with all different digits. The time from 19:58:47 to 20:13:45 is 13 sec + 1 min + 13 min + 45 sec or 14 min + 58 sec = 14 × 60 sec + 58 sec = 898 seconds.

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